

Q1

$$F = 100 \text{ kmol}$$

$$z_a = 0.6$$

$$z_T = 0.25$$

$$z_x = 0.15$$

$$P = 1 \text{ atm} = 760 \text{ mm of Hg}$$

} given in question

Using Antoine Equation

$$\log_{10} P = A - \frac{B}{C+T}$$

$P \rightarrow$ is in mm of Hg
 $T \rightarrow$ is in $^{\circ}\text{C}$

For Benzene: $A = 6.90565$

$$B = 1211.033$$

$$C = 220.79$$

For Toluene: $A = 6.9546$

$$B = 1344.8$$

$$C = 219.482$$

For O-Xylene: $A = 6.99891$

$$B = 1474.67$$

$$C = 213.686$$

For Ideal gases: $k = \frac{P_i}{P}$ where P_i : saturated partial pressure

$$k_B = 1.77696; k_T = 0.732; k_x = 0.2612$$

The 'P' value calculated through antoine equation is directly used in the code
 'P' matrix is used

The Rachford - Rice model states that

$$0 = \sum_{i=1}^3 \frac{z_i (1 - K_i)}{1 + \theta (K_i - 1)} \quad \text{where } \theta = \frac{D}{F}$$

a) From code we can get

$$@ T = 100^{\circ}\text{C}, \theta = 0.6743$$

$$\text{Distillate flow rate (D)} = 67.43 \text{ kmol/hr}$$

$$\text{Bottom flow rate (B)} = 32.57 \text{ kmol/hr}$$

} Total flow rate = 100 kmol/hr

Distillate composition

$$y_B = 0.6989$$

$$y_T = 0.2230$$

$$y_X = 0.0782$$

Bottom composition

$$x_B = 0.3953$$

$$x_T = 0.3060$$

$$x_X = 0.2987$$

Iteration | Error

```
-----
1      | 0.0043289495679043055
2      | 1.4903948996058958e-05
3      | 1.7946455432849007e-10
```

Converged Value of phi: 0.6743140454394437

$\theta = \phi$ (same)

Top Flow Rate (D): 67.43 kmol/hr

Bottom Flow Rate (B): 32.57 kmol/hr

Mole fraction composition of the top product:

Benzene: 0.6989 %mol

Toluene: 0.2230 %mol

o-Xylene: 0.0782 %mol

Mole fraction composition of the bottom product:

Benzene: 0.3953 %mol

Toluene: 0.3060 %mol

o-Xylene: 0.2987 %mol

b) Code output

```
For Temperature: 100.00 °C
Iteration | Error
-----
1      | 0.5682101501949811
2      | 0.006133622675591477
3      | 2.9726717136746572e-05
4      | 7.139920876753081e-10

Converged Value of phi for 100.00 °C: 0.67431
```

Top Flow Rate (D): 67.43 kmol/hr
Bottom Flow Rate (D): 32.57 kmol/hr

Mole fraction composition of the top product:

Benzene: 0.6989 %mol

Toluene: 0.2230 %mol

o-Xylene: 0.0782 %mol

Mole fraction composition of the bottom product:

Benzene: 0.3953 %mol

Toluene: 0.3060 %mol

o-Xylene: 0.2987 %mol

Recovery of Benzene: 1.1647829718116605

For Temperature: 100.78 °C

Iteration | Error

```
-----
1      | 0.506984389738601
2      | 0.022911855126422576
3      | 0.0004996506061670791
4      | 2.180278367847066e-07
5      | 4.141131881851834e-14
```

Converged Value of phi for 100.78 °C: 0.71357

Top Flow Rate (D): 71.36 kmol/hr
Bottom Flow Rate (D): 28.64 kmol/hr

Mole fraction composition of the top product:

Benzene: 0.6879 %mol

Toluene: 0.2278 %mol

o-Xylene: 0.0843 %mol

Mole fraction composition of the bottom product:

Benzene: 0.3810 %mol

Toluene: 0.3053 %mol

o-Xylene: 0.3137 %mol

Recovery of Benzene: 1.1465079871115127

For Temperature: 101.56 °C

Iteration | Error

```
-----
1      | 0.40991158824233664
2      | 0.03728072434874963
3      | 0.0015119805607416437
4      | 2.153077158006944e-06
5      | 4.340749981679437e-12
```

Converged Value of phi for 101.56 °C: 0.75112

Top Flow Rate (D): 75.11 kmol/hr
Bottom Flow Rate (D): 24.89 kmol/hr

Mole fraction composition of the top product:

Benzene: 0.6771 %mol

Toluene: 0.2321 %mol

o-Xylene: 0.0908 %mol

Mole fraction composition of the bottom product:

Benzene: 0.3673 %mol

Toluene: 0.3039 %mol

o-Xylene: 0.3288 %mol

Recovery of Benzene: 1.1285263977435862

For Temperature: 102.33 °C

Iteration | Error

```
-----
1      | 0.3247011538065109
2      | 0.036021942341666
3      | 0.001507091287884399
4      | 2.287894386832079e-06
5      | 5.241806988465214e-12
```

Converged Value of phi for 102.33 °C: 0.78717

Top Flow Rate (D): 78.72 kmol/hr
Bottom Flow Rate (D): 21.28 kmol/hr

Mole fraction composition of the top product:

Benzene: 0.6665 %mol

Toluene: 0.2360 %mol

o-Xylene: 0.0975 %mol

Mole fraction composition of the bottom product:

Benzene: 0.3541 %mol

Toluene: 0.3018 %mol

o-Xylene: 0.3441 %mol

Recovery of Benzene: 1.1108247871737278

```
For Temperature: 103.11 °C
Iteration | Error
-----
1          | 0.2524443509016995
2          | 0.029460936131007087
3          | 0.0010462324100130038
4          | 1.1718519378201009e-06
5          | 1.4639400802707314e-12

Converged Value of phi for 103.11 °C: 0.82194

Top Flow Rate (D): 82.19 kmol/hr
Botton Flow Rate (D): 17.81 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6560 %mol
Toluene: 0.2394 %mol
o-Xylene: 0.1046 %mol

Mole fraction composition of the bottom product:
Benzene: 0.3414 %mol
Toluene: 0.2991 %mol
o-Xylene: 0.3595 %mol

Recovery of Benzene: 1.093385315394531
-----
```

```
For Temperature: 103.89 °C
Iteration | Error
-----
1          | 0.19908498228038063
2          | 0.022832461886473365
3          | 0.0006485880002458089
4          | 4.766344527951105e-07
5          | 2.566835632933362e-13

Converged Value of phi for 103.89 °C: 0.85560

Top Flow Rate (D): 85.56 kmol/hr
Botton Flow Rate (D): 14.44 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6457 %mol
Toluene: 0.2423 %mol
o-Xylene: 0.1120 %mol

Mole fraction composition of the bottom product:
Benzene: 0.3291 %mol
Toluene: 0.2958 %mol
o-Xylene: 0.3750 %mol

Recovery of Benzene: 1.0761865463501827
-----
```

```
For Temperature: 106.22 °C
Iteration | Error
-----
1          | 0.043836670463878846
2          | 0.002117194881927742
3          | 5.937166427383289e-06
4          | 4.62676563728337e-11

Converged Value of phi for 106.22 °C: 0.95171

Top Flow Rate (D): 95.17 kmol/hr
Botton Flow Rate (D): 4.83 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6155 %mol
Toluene: 0.2483 %mol
o-Xylene: 0.1362 %mol

Mole fraction composition of the bottom product:
Benzene: 0.2951 %mol
Toluene: 0.2831 %mol
o-Xylene: 0.4218 %mol

Recovery of Benzene: 1.0257832509580382
-----
```

```
For Temperature: 107.00 °C
Iteration | Error
-----
1          | 0.0026490135560242045
2          | 9.496858777380268e-06
3          | 1.234710111930326e-10

Converged Value of phi for 107.00 °C: 0.98264

Top Flow Rate (D): 98.26 kmol/hr
Botton Flow Rate (D): 1.74 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6056 %mol
Toluene: 0.2495 %mol
o-Xylene: 0.1449 %mol

Mole fraction composition of the bottom product:
Benzene: 0.2845 %mol
Toluene: 0.2781 %mol
o-Xylene: 0.4374 %mol

Recovery of Benzene: 1.0092889449558247
-----
```

```
For Temperature: 104.67 °C
Iteration | Error
-----
1          | 0.13147464351357063
2          | 0.012917057847219526
3          | 0.00021016329037537584
4          | 5.2727185728684844e-08
5          | 3.219646771412954e-15

Converged Value of phi for 104.67 °C: 0.88835

Top Flow Rate (D): 88.83 kmol/hr
Botton Flow Rate (D): 11.17 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6355 %mol
Toluene: 0.2447 %mol
o-Xylene: 0.1198 %mol

Mole fraction composition of the bottom product:
Benzene: 0.3174 %mol
Toluene: 0.2920 %mol
o-Xylene: 0.3906 %mol

Recovery of Benzene: 1.0592042779243944
-----
```

```
For Temperature: 105.44 °C
Iteration | Error
-----
1          | 0.08728289206756146
2          | 0.006889397809749065
3          | 6.127518697995882e-05
4          | 4.70835581634077e-09
5          | 1.1102230246251565e-16

Converged Value of phi for 105.44 °C: 0.92033

Top Flow Rate (D): 92.03 kmol/hr
Botton Flow Rate (D): 7.97 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6254 %mol
Toluene: 0.2467 %mol
o-Xylene: 0.1278 %mol

Mole fraction composition of the bottom product:
Benzene: 0.3060 %mol
Toluene: 0.2878 %mol
o-Xylene: 0.4062 %mol

Recovery of Benzene: 1.042412328964906
-----
```

c)

```

For Temperature: 100.00 °C
Iteration | Error
-----
1 | 0.5682101501949811
2 | 0.006133622675591477
3 | 2.9726717136746572e-05
4 | 7.139920876753081e-10

Converged Value of phi for 100.00 °C: 0.67431

Top Flow Rate (D): 67.43 kmol/hr
Bottom Flow Rate (D): 32.57 kmol/hr

Mole fraction composition of the top product:
Benzene: 0.6989 %mol
Toluene: 0.2230 %mol
o-Xylene: 0.0782 %mol

Mole fraction composition of the bottom product:
Benzene: 0.3953 %mol
Toluene: 0.3060 %mol
o-Xylene: 0.2987 %mol

Recovery of Benzene: 1.1647829718116605

```

Maximum recovery = 1.1648
at $T = 100^\circ\text{C}$

d) α is taken to be in proportion to bottom product which is recycled in the code.
So, clearly for higher α value, the benzene recovered is more

Q2

$$F = 1000 \text{ kmol/hr}$$

$$Z_E = 0.2$$

$$T_E = 98^\circ\text{C}$$

$$Z_M = 0.6$$

$$T_M = 65^\circ\text{C}$$

$$Z_P = 0.2$$

$$T_P = 97^\circ\text{C}$$

} given

$$\text{No of stages} = 3$$

$$P_{\text{operating}} = 1 \text{ atm}$$

$$\text{Reflux (external)} = 2000 \text{ kmol/hr}$$

$$D = 600 \text{ kmol/hr}$$

Assuming a linear Temperature profile from 65°C to 97°C throughout the distillation column

The converged temperature profile in K is :

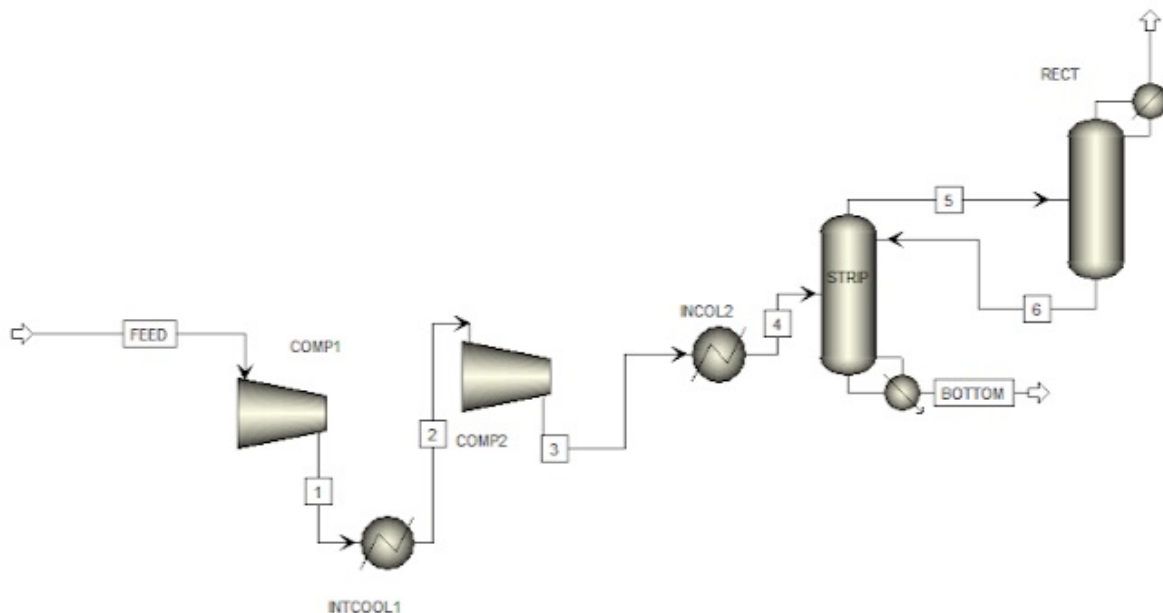
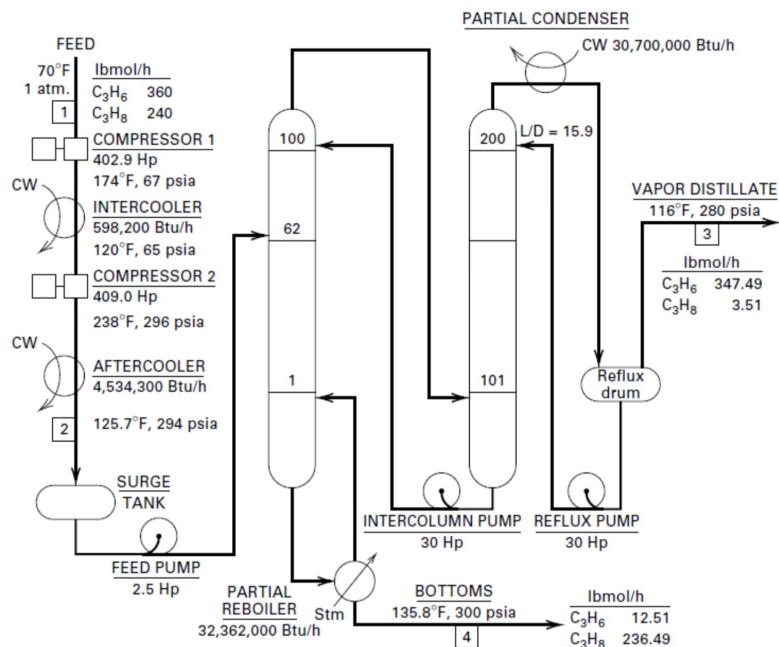
Iteration | Error

1	0.001434674761852986
2	0.00025486155038104163
3	4.849289210573447e-05
4	8.237975414984897e-06
5	1.3175689068732306e-06
6	2.0511622178854202e-07

Converged Temperature profile - [340.14047722 341.99953841 344.85874807 348.47575624 354.26167834]

b) From code, for $m = 10$, optimum feed location = 6

Q3



Heat and Material Balance Table

Stream ID		1	2	3	4	5	6	BOTTOM	FEED	VAPDIST
Temperature	C	98.4	48.9	131.8	52.1	49.9	50.1	58.1	21.1	47.1
Pressure	bar	4.619	4.482	20.408	20.271	19.995	20.119	20.677	1.013	19.305
Vapor Frac		1.000	1.000	1.000	0.815	1.000	0.000	0.000	1.000	1.000
Mole Flow	kmol/hr	272.155	272.155	272.155	272.155	3094.357	2935.146	112.945	272.155	159.213
Mass Flow	kg/hr	11671.927	11671.927	11671.927	11671.927	131373.105	124671.996	4970.819	11671.927	6701.933
Volume Flow	cum/hr	1750.176	1533.081	388.437	210.536	2919.639	272.489	11.494	6465.341	156.748
Enthalpy	Gcal/hr	-1.600	-1.851	-1.489	-2.107	-2.697	-11.672	-2.994	-1.957	0.709
Mole Flow	kmol/hr									
PROPY-01		163.293	163.293	163.293	163.293	2518.645	2360.134	4.782	163.293	158.146
PROPA-01		108.862	108.862	108.862	108.862	575.712	575.012	108.162	108.862	1.067
Mole Frac										
PROPY-01		0.600	0.600	0.600	0.600	0.814	0.804	0.042	0.600	0.993
PROPA-01		0.400	0.400	0.400	0.400	0.186	0.196	0.958	0.400	0.007

[illegible][illegible]